

account by [26] to apply exponential schemes to the SWE on the plane. In this work, we overcome this issue by including the Coriolis effects into the nonlinear term, *i.e.*, we consider as linear and nonlinear terms respectively $\mathbf{L} := \mathbf{L}_G$ and $\mathbf{N} := \mathbf{L}_C + \mathbf{N}_A + \mathbf{N}_R + \mathbf{b}$. Alternative splitting between linear and nonlinear terms is also done in operational models, *e.g.*, the Integrated Forecast System of the European Centre for Medium-Range Weather Forecast (IFS-ECMWF) [11], in which the linear Coriolis term is incorporated into the nonlinear advection.

Since we consider a spectral discretization using spherical harmonics, the time integration (3.1) is performed for the spectral coefficients $\mathbf{U}_{m,n} := (\Phi_{m,n}, \xi_{m,n}, \delta_{m,n})^T$ with zonal and total wavenumbers m and n , respectively. In the spectral space, $\mathbf{L}\mathbf{U}$ becomes, for each wavenumber pair (m, n) , $\mathcal{L}\mathbf{U}_{n,m}$, where $\mathcal{L}$ is the symbol matrix of $\mathbf{L} = \mathbf{L}_G$, given by:

$$\mathcal{L} = \begin{pmatrix} 0 & 0 & -\bar{\Phi} \\ 0 & 0 & 0 \\ -D_n & 0 & 0 \end{pmatrix} \quad (3.9)$$

where $D_n := -n(n+1)/a^2$ and a is the Earth's radius. The eigenvalues of the symbol matrix, which are the diagonal elements of Λ in (3.8), read $-\sqrt{-D_n\bar{\Phi}}$, 0 and $\sqrt{-D_n\bar{\Phi}}$.

An alternative approach to obtain a diagonalizable linear operator would be to use spectral methods considering Hough harmonics [20], which are eigenmodes of $\mathbf{L}_G + \mathbf{L}_C$, as basis functions. This is done by [42] to solve the SWE on the rotating sphere using exponential methods. However, implementing the Hough decomposition is much less straightforward and computationally more expensive than the spherical harmonics one, for which there are efficient available open-source libraries, *e.g.*, SHTns [31]; therefore, we consider the latter in this work.

In terms of computational efficiency, it is important to notice that, for a fixed problem (*i.e.*, for a fixed linear operator $\mathbf{L}$), the φ_k functions evaluated at $\Delta t \mathbf{L}$, as used in the ETDRK schemes, depend only on the time step. Therefore, if the time step is constant during a numerical simulation, these functions can be precomputed. This may lead to a significant reduction in terms of computing time, since the φ_k functions need to be computed for every pair of wavenumbers (m, n) .

3.2. SL-SI-SETTLS

Semi-Lagrangian (SL) schemes are spatio-temporal discretization methods for advective PDEs that combine the Eulerian and Lagrangian approaches. The former is based on fixed spatial grids and usually has restrictions on the time step size imposed by Courant-Friedrichs-Lewy (CFL) stability conditions, and the latter consists of following fluid particles along their trajectories, which allows the use of larger time steps but is more complex in terms of implementation compared to fixed grid approaches. The SL approach follows the Lagrangian trajectories only along individual time steps: the trajectories arriving at each point of the fixed spatial grid at time t_{n+1} are estimated, and a spatial interpolation procedure is performed to retrieve the solution at the departure points at time t_n . This simplified Lagrangian approach, which still allows the use of relatively large time step sizes [37], explains the popularity of SL methods, which are used by several weather and climate agencies [23].

An important example of SL methods is the semi-Lagrangian semi-implicit SETTLS (SL-SI-SETTLS), a second-order scheme used in the IFS-ECMWF [11]. This scheme uses a semi-implicit (Crank-Nicolson) approach for the linear terms and the Stable Extrapolation Two-Time-Level Scheme (SETTLS), proposed by [17], for estimating the Lagrangian trajectories $(t, \mathbf{x}(t))$ at each time step $[t_n, t_{n+1}]$. The trajectories between the departure point $(t_n, \mathbf{x}_d)$ and the arrival (grid)

point $(t_{n_1}, \mathbf{x}_j)$ are considered to be linear and determined by the velocity field $\mathbf{v}$ at its midpoint $(t_{n+1/2}, \mathbf{x}_m)$, where $t_{n+1/2} := t_n + \Delta t/2$. This velocity field, on its turn, is approximated by a linear interpolation between $\mathbf{v}(t_n, \mathbf{x}_d)$ and $\mathbf{v}(t_{n+1}, \mathbf{x}_j)$, with this last one being estimated via a linear extrapolation using the two previous time steps:

$$\mathbf{x}_j - \mathbf{x}_d = \Delta t \mathbf{v}(t_{n+1/2}, \mathbf{x}_m) = \frac{\Delta t}{2} [2\mathbf{v}(t_n, \mathbf{x}_d) - \mathbf{v}(t_{n-1}, \mathbf{x}_d) + \mathbf{v}(t_n, \mathbf{x}_j)]. \quad (3.10)$$

For each grid point $\mathbf{x}_j$, Eq. (3.10) is solved iteratively to obtain the departure point $\mathbf{x}_d$. Then, SL-SI-SETTLS applied to (2.1) reads

$$\frac{\mathbf{U}^{n+1} - \mathbf{U}_*^n}{\Delta t} = \frac{1}{2} (\mathbf{L}\mathbf{U}^{n+1} + (\mathbf{L}\mathbf{U}^n)_*) + \frac{1}{2} ([2\tilde{\mathbf{N}}(\mathbf{U}^n) - \tilde{\mathbf{N}}(\mathbf{U}^{n-1})]_* + \tilde{\mathbf{N}}(\mathbf{U}^n)) \quad (3.11)$$

where quantities denoted by the subscript $*$ are interpolated to the departure point $\mathbf{x}_d$ at time instant t_n . As in the IFS-ECMWF model, we consider the linear term to include only gravity-related processes, *i.e.*, $\mathbf{L} := \mathbf{L}_G$; also, we consider both the Coriolis- and the bathymetry-related linear terms to be treated as nonlinear ones, such that $\tilde{\mathbf{N}} := \mathbf{N}_R + \mathbf{L}_C + \mathbf{b}$.

3.3. SL-ETDRK

A semi-Lagrangian version of the ETDRK schemes is proposed by [26], aiming to tackle problems of the form

$$\frac{D\mathbf{U}}{Dt} = \mathbf{L}\mathbf{U} + \tilde{\mathbf{N}}(\mathbf{U}), \quad \mathbf{U}(t=0) = \mathbf{U}_0, \quad (3.12)$$

where $D/Dt := \partial/\partial t + \mathbf{V} \cdot \nabla$ is the material derivative, $\mathbf{V}$ is a velocity field, $\tilde{\mathbf{N}}$ is a nonlinear term not containing the nonlinear advection $\mathbf{V} \cdot \nabla$ and $\mathbf{U} = \mathbf{U}(t, \mathbf{x}(t))$ is the solution defined on the Lagrangian trajectories. The schemes are derived analogously to the Eulerian case (Section 3.1): the exact solution of (3.12) reads

$$\mathbf{U}(t_{n+1}, \mathbf{x}(t_{n+1})) = P_n^{-1}(t_{n+1})\mathbf{U}(t_n, \mathbf{x}(t_n)) + P_n^{-1}(t_{n+1}) \int_{t_n}^{t_{n+1}} P_n(s) \tilde{\mathbf{N}}(\mathbf{U}(s, \mathbf{x}(s))) ds, \quad (3.13)$$

where $P_n(t)$ is the solution of an integration factor problem along the Lagrangian trajectory in $[t_n, t_{n+1}]$, assumed to exist and be invertible for all times. If $\mathbf{L}$ depends on the space, then it depends on the specific Lagrangian trajectory $(t, \mathbf{x}(t))$. Note that this is the case of the SWE on the rotating sphere (2.1), since $\mathbf{L}$ is a differential operator; if $\mathbf{L}$ were constant, then the derivative (*i.e.*, the application of $\mathbf{L}$) would be constant along the trajectory. Let $\mathbf{L}_{\mathbf{x}(t)}$ denote the trajectory-dependent linear operator. If it satisfies $\mathbf{L}_{\mathbf{x}(t)}\mathbf{L}_{\mathbf{x}(s)} = \mathbf{L}_{\mathbf{x}(s)}\mathbf{L}_{\mathbf{x}(t)}$ for every t, s , then P_n reads

$$P_n(t) = e^{-\int_{t_n}^t \mathbf{L}_{\mathbf{x}(s)} ds}. \quad (3.14)$$

In [26], the linear term is assumed to be constant along each trajectory and each time step, such that the integration factor assumes the same form as in the Eulerian exponential schemes presented in Section 3.1. The schemes analogous to ETD1RK and ETD2RK were named respectively SL-ETD1RK and SL-ETD2RK, in which first- and second-order approximations are proposed to the nonlinear term. However, as will be shown in Section 4, the approximation of the linear term is

actually a first-order one, such that both schemes are globally first-order. Therefore, we rename these methods respectively as *SE11* and *SE12*, where the first and second numbers refer to the orders of approximation of the linear and nonlinear terms, respectively. These schemes read

$$\mathbf{U}_{\text{SE11}}^{n+1} = \varphi_0(\Delta t \mathbf{L}) \left[\mathbf{U}^n + \Delta t \psi_1(\Delta t \mathbf{L}) \tilde{\mathbf{N}}(\mathbf{U}^n) \right]_*^n, \quad (3.15a)$$

$$\mathbf{U}_{\text{SE12}}^{n+1} = \mathbf{U}_{\text{SE11}}^{n+1} + \Delta t \varphi_0(\Delta t \mathbf{L}) \left[\psi_2(\Delta t \mathbf{L}) \tilde{\mathbf{N}}(\mathbf{U}_{\text{SE11}}^{n+1}) - \left(\psi_2(\Delta t \mathbf{L}) \tilde{\mathbf{N}}(\mathbf{U}^n) \right)_*^n \right], \quad (3.15b)$$

where the functions ψ_k also satisfy the property (3.8) and are defined by

$$\psi_k(z) := (-1)^{k+1} \varphi_k(-z) + \sum_{l=1}^{k-1} \varphi_l(-z), \quad k \geq 1. \quad (3.16)$$

The semi-Lagrangian trajectories are estimated as in SL-SI-SETTLS, by solving (3.10) iteratively. Finally, in (3.15a) and (3.15b), the nonlinear term does not contain the advection effects, which are treated by the SL approach. Thus, the only nonlinear term included in $\tilde{\mathbf{N}}$ is $\mathbf{N}_{\mathbf{R}}$, such that $\tilde{\mathbf{N}} = \mathbf{N}_{\mathbf{R}} + \mathbf{L}_{\mathbf{C}} + \mathbf{b}$. As in the Eulerian exponential schemes, the treatment of the Coriolis effects as a nonlinear term allows us to compute the matrix exponentials accurately using (3.8), since $\mathbf{L} := \mathbf{L}_{\mathbf{G}}$. We remark that, in the SL framework, the Coriolis term could also be dealt with by considering it as an advected quantity [40].

4. SOURCE OF THE FIRST ORDER ACCURACY OF SE12

We demonstrate in the following paragraphs that the term $\varphi_0(\Delta t \mathbf{L})[\mathbf{U}^n]_*^n$ is responsible for the first-order accuracy of SE12, despite a second-order approximation to the nonlinear term. More specifically, it is due to the approximation of the integral in the solution (3.14) of the integration factor problem by considering the integrand to be constant in $[t_n, t_{n+1}]$. For that, we compute the truncation error of the scheme

$$\mathbf{U}^{n+1} = \varphi_0(\Delta t \mathbf{L})[\mathbf{U}^n]_*^n \quad (4.1)$$

approximating the linear PDE,

$$\frac{D\mathbf{u}}{Dt} = L\mathbf{u}, \quad \frac{D}{Dt} := \frac{\partial}{\partial t} + v \frac{\partial}{\partial x}, \quad (4.2)$$

where $v = v(t, x)$ is a velocity field and $L = L(x)$ is a space-dependent linear operator. For the sake of conciseness, we restrict ourselves to the one-dimensional case in space, with a homogeneous spatial discretization with mesh size Δx , and we assume that all quantities are scalars. In Section 5, in which we propose a second-order method, some remarks are necessary in the case of systems of PDEs.

Along each Lagrangian trajectory $(t, x(t))$, (4.2) reduces to the ODE

$$\frac{d\mathbf{u}}{dt}(t, x(t)) = L(x(t))\mathbf{u}(t, x(t)). \quad (4.3)$$

Consider a spatial discretization of (4.3), with $P + 1$ points $x_0, \dots, x_P$ and homogeneous step Δx . Let $\Gamma_j^{n+1}(t) := (t, x^j(t))$ be the Lagrangian trajectory in $[t_n, t_{n+1}]$, arriving at the grid point x_j at time t_{n+1} , such that $x^j(t_{n+1}) = x_j$. Under this notation, we also define $u_{e_j}^n := u(t_n, x_{e_j}) :=$